

Technology Stack

Date	30 September 2026
Team ID	SWTID-2026-1418
Project Name	fit buddy
Maximum Marks	2 Marks

Define Your Technology Stack:

Identify the programming languages, frameworks, databases, front-end tools, back-end tools, and APIs that you will use to build your solution. A well-chosen technology stack ensures that your application is scalable, maintainable, and aligned with your project requirements.

Fill out the table below to detail the specific technologies chosen for each component of your architecture and provide a brief justification for your choice.

Technology Stack Details

S.No	Architecture Component / Layer	Technology Chosen	Justification / Purpose
1	Frontend / Client-Side	React.js, HTML, CSS, JavaScript	React provides reusable components and a responsive, user-friendly interface for the Fit buddy application.
2	Backend / Server-Side	Python + FastAPI	FastAPI is lightweight and fast, making it suitable for building APIs and connecting the frontend with AI and database services.
3	Database / Data Storage	PostgreSQL	PostgreSQL provides reliable relational data storage for user profiles, fitness goals, workout plans, and progress records.
4	Cloud / Hosting / Deployment	Docker + Render	Docker provides a consistent deployment environment, while Render makes it easy to deploy and scale the application.
5	Version Control & CI/CD	Git + GitHub + GitHub Actions	Git and GitHub support team collaboration and version control, while GitHub Actions can automate testing and deployment.
6	Third-Party APIs / Other Tools	OpenAI API	The OpenAI API enables AI-powered personalized fitness recommendations, workout guidance, and conversational assistance.